Algorithms for Relational Database Schema Design

Relational Synthesis into 3NF with Dependency Preservation (Relational Synthesis Algorithm)

- **Input: A universal relation R and a set of functional dependencies F on the attributes of R.**

1. Find a minimal cover G for F;

2. For each left-hand-side X of a functional dependency that appears in G,

create a relation schema in D with attributes {X υ {A1} υ {A2} ... υ {Ak}},

where X → A1, X → A2, ..., X → Ak are the only dependencies in G with X as left-hand-side (X is the key of this relation) ;

3. Place any remaining attributes (that have not been placed in any relation) in a single relation schema to ensure the attribute preservation property.

Claim: Every relation schema created by Algorithm is in 3NF.

U(Essn, Pno, Esal, Ephone, Dno, Pname, Plocation)

FD:

Essn→ Esal, Ephone, Dno

Pno→Pname, Plocation

Essn, Pno → Esal, Ephone, Dno, Pname, Plocation

Minimal Cover:

Essn→ Esal, Ephone, Dno

Pno→Pname, Plocation

Decomposed relations:

R1(<u>Essn</u>, Esal, Ephone, Dno)

R2(<u>Pno</u>,Pname, Plocation)

Algorithms for Relational Database Schema Design

Relational Decomposition into BCNF with Lossless (non-additive) join property

- **Input: A universal relation R and a set of functional dependencies F on the attributes of R.**

1. Set D := {R};

2. While there is a relation schema Q in D that is not in BCNF

do {

choose a relation schema Q in D that is not in BCNF;

find a functional dependency X → Y in Q that violates BCNF;

replace Q in D by two relation schemas (Q - Y) and (X υ Y);

};

Assumption: No null values are allowed for the join attributes.

DB(Patno,PatName,appNo,time,doctor)
Patno -> PatName
Patno -> Time,doctor
Time -> appNo
appNo-> Patno

2NF AND 3NF

DB(<u>Patno</u>,appNo,Time,doctor, PatName)

BCNF

Q=DB(<u>Patno</u>,appNo,Time,doctor, PatName)
Time -> appNo (X→Y, violates BCNF)
Replace Q into: Q-Y AND XUY
D1(<u>Patno</u>,time,doctor, PatName)
D2(<u>Time</u>, appNo)

General Procedure for achieving BCNF when a relation fails BCNF

Let R be the relation not in BCNF, let X be a subset-of R, and let $X \rightarrow A$ be the FD that causes a violation of BCNF. Then R may be decomposed into two relations:

(i) $R - A$ and (ii) $X \cup A$.

If either $R - A$ or $X \cup A$. is not in BCNF, repeat the process.

Note that the f.d. that violated BCNF in TEACH(Instructor, Student, Course)

was Instructor $\rightarrow$Course. Hence its BCNF decomposition would be (TEACH – COURSE) and (Instructor $\cup$ Course), which gives

the relations: (Instructor, Student) and (Instructor, Course) that we obtained before in decomposition D3.

Finding a Key K for R Given a set F of Functional Dependencies

Input:

A universal relation R and a set of functional dependencies F on attributes of R.

Method:

1. Set K:= R.

2. For each attribute A in K
   {compute $(K - A)^+$ with respect to F;
   if $(K - A)^+$ contains all the attributes in R, then set K := K - {A}
   };

Let R = (A, B, C, D, E) be a relation scheme with the following dependencies-

$AB \rightarrow C$

$C \rightarrow D$

$B \rightarrow E$

Algorithms for Relational Database Schema Design

Relational Synthesis into 3NF with Dependency Preservation and Lossless (Non-Additive) Join Property

- **Input: A universal relation R and a set of functional dependencies F on the attributes of R.**

1. Find a minimal cover G for F.

2. For each left-hand-side X of a functional dependency that appears in G,

create a relation schema in D with attributes {X ∪ {A1} ∪ {A2} ... ∪ {Ak}},

where X → A1, X → A2, ..., X –>Ak are the only dependencies in G with X as left-hand-side (X is the key of this relation).

3. If none of the relation schemas in D contains a key of R, then create one more relation schema in D that contains attributes that form a key of R. *(Use Algorithm to find the key of R)*

4. Eliminate redundant relations from the resulting set of relations in the relational database schema.

Example

FD:

$\{P \rightarrow LCA, LC \rightarrow AP, A \rightarrow C\}$

Minimal Cover: $P \rightarrow LC, LC \rightarrow AP, A \rightarrow C$

Design X: R1(<u>P</u>,L,C), R2(<u>L,C</u>, A, P) and R3(<u>A</u>,C)

R3 is subsumed by R2, R1 by R2 //eliminating redundant relations

Final Design: R2(L,C,A,P)

F: $\{P \rightarrow LCA, LC \rightarrow AP, A \rightarrow C\}$

$LC \rightarrow P$, $P \rightarrow A$ => $LC \rightarrow A$
$P \rightarrow A$, $A \rightarrow C$ => $P \rightarrow C$

MC:$\{P \rightarrow LA, LC \rightarrow P, A \text{->} C\}$

Alternate Design: S1(<u>P</u>,A,L),S2(<u>L,C</u>,P) AND S3(<u>A</u>,C)

Problems with NULL values and Dangling Tuples

Problems with NULL values

when some tuples have NULL values for attributes that will be used to join individual relations in the decomposition that may lead to incomplete results.

E.g., see Figure 15.2(a), where two relations EMPLOYEE and DEPARTMENT are shown. The last two employee tuples—'Berger' and 'Benitez'—represent newly hired employees who have not yet been assigned to a department (assume that this does not violate any integrity constraints).

If we want to retrieve a list of (Ename, Dname) values for all the employees. If we apply the NATURAL JOIN operation on EMPLOYEE and DEPARTMENT (Figure 15.2(b)), the two aforementioned tuples will *not* appear in the result.

In such cases, LEFT OUTER JOIN may be used. The result is shown in Figure 15.2 (c).

(a)

EMPLOYEE

Ename	Ssn	Bdate	Address	Dnum
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4
Narayan, Ramesh K.	666884444	1962-09-15	975 Fire Oak, Humble, TX	5
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1
Berger, Anders C.	999775555	1965-04-26	6530 Braes, Bellaire, TX	NULL
Benitez, Carlos M.	888664444	1963-01-09	7654 Beech, Houston, TX	NULL

DEPARTMENT

Dname	Dnum	Dmgr_ssn
Research	5	333445555
Administration	4	987654321
Headquarters	1	888665555

Figure 15.2
Issues with NULL-value joins. (a) Some EMPLOYEE tuples have NULL for the join attribute Dnum.

Problems with Null Values and Dangling Tuples

(b)

Ename	Ssn	Bdate	Address	Dnum	Dname	Dmgr_ssn
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5	Research	333445555
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5	Research	333445555
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4	Administration	987654321
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4	Administration	987654321
Narayan, Ramesh K.	666884444	1962-09-15	975 Fire Oak, Humble, TX	5	Research	333445555
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5	Research	333445555
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4	Administration	987654321
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1	Headquarters	888665555

(c)

Ename	Ssn	Bdate	Address	Dnum	Dname	Dmgr_ssn
Smith, John B.	123456789	1965-01-09	731 Fondren, Houston, TX	5	Research	333445555
Wong, Franklin T.	333445555	1955-12-08	638 Voss, Houston, TX	5	Research	333445555
Zelaya, Alicia J.	999887777	1968-07-19	3321 Castle, Spring, TX	4	Administration	987654321
Wallace, Jennifer S.	987654321	1941-06-20	291 Berry, Bellaire, TX	4	Administration	987654321
Narayan, Ramesh K.	666884444	1962-09-15	975 Fire Oak, Humble, TX	5	Research	333445555
English, Joyce A.	453453453	1972-07-31	5631 Rice, Houston, TX	5	Research	333445555
Jabbar, Ahmad V.	987987987	1969-03-29	980 Dallas, Houston, TX	4	Administration	987654321
Borg, James E.	888665555	1937-11-10	450 Stone, Houston, TX	1	Headquarters	888665555
Berger, Anders C.	999775555	1965-04-26	6530 Braes, Bellaire, TX	NULL	NULL	NULL
Benitez, Carlos M.	888665555	1963-01-09	7654 Beech, Houston, TX	NULL	NULL	NULL

Figure 15.2
Issues with NULL-value joins.
(b) Result of applying NATURAL JOIN to the EMPLOYEE and DEPARTMENT relations.
(c) Result of applying LEFT OUTER JOIN to EMPLOYEE and DEPARTMENT

About Normalization Algorithms

Discussion of Normalization Algorithms:

Problems:

- The database designer must first specify *all* the relevant functional dependencies among the database attributes.
- These algorithms are *not deterministic* in general.
- It is not always possible to find a decomposition into relation schemas that preserves dependencies and allows each relation schema in the decomposition to be in BCNF.

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Algorithms for Relational Database Schema Design

Table 11.1
Summary of the Algorithms Discussed in Sections 11.1 and 11.2

Algorithm	Input	Output	Properties/Purpose	Remarks
11.1	A decomposition *D* of *R* and a set *F* of functional dependencies	Boolean result: yes or no for nonadditive join property	Testing for nonadditive join decomposition	See a simpler test in Section 11.1.4 for binary decompositions
11.2	Set of functional dependencies *F*	A set of relations in 3NF	Dependency preservation	No guarantee of satisfying lossless join property
11.3	Set of functional dependencies *F*	A set of relations in BCNF	Nonadditive join decomposition	No guarantee of dependency preservation
11.4	Set of functional dependencies *F*	A set of relations in 3NF	Nonadditive join *and* dependency-preserving decomposition	May not achieve BCNF, but achieves *all* desirable properties and 3NF
11.4a	Relation schema *R* with a set of functional dependencies *F*	Key K of *R*	To find a key K (that is a subset of *R*)	The entire relation *R* is always a default superkey

Thank you